

Resume Scoring System – Technical Writeup

Introduction

The Resume Scoring System is a hybrid AI-based application designed to evaluate how effectively a candidate’s resume matches a given job description (JD). The system generates a fit score (0–100) along with explainable insights such as matched skills, missing requirements, improvement suggestions, and uncertainty warnings.

The project was designed with a strong focus on explainability, modularity, and lightweight inference. Instead of relying entirely on a single black-box LLM call, the system combines deterministic rule-based techniques with lightweight NLP methods to ensure transparent and reproducible scoring.

1. Schema Design

The system converts resumes and job descriptions into a structured internal schema for consistent evaluation.

```
@dataclass
```

```
class ParsedDocument:
```

```
    raw_text: str
```

```
    skills: List[str]
```

```
    years_experience: float
```

```
    education: List[str]
```

```
    responsibilities: List[str]
```

```
    evidence: Dict[str, List[str]]
```

```
    warnings: List[str]
```

Schema Rationale

Field	Purpose
skills	Extracted technical/domain skills
years_experience	Relevant experience identified
education	Degree-level detection
responsibilities	Work responsibility extraction
evidence	Skill → supporting sentence mapping
warnings	Suspicious or low-quality input detection

The evidence field improves explainability by showing why a skill was matched, while the warnings field helps detect adversarial or low-quality resumes.

2. Scoring Logic

The final score is calculated using weighted category scores.

Category	Weight
Required Skills	50%
Responsibilities	15%
Experience	15%
Education	10%
Semantic Similarity	10%

Final Score Formula

Final Score= $\sum$ (Category Score $\times$ Weight)

Additional bonus points are applied for exceptionally strong matches, while warning penalties reduce scores for suspicious inputs such as keyword stuffing or repetitive content.

Category Scoring

Skills Score

Calculated using skill overlap percentage between the resume and JD.

Responsibilities Score

Uses TF-IDF vectorization and cosine similarity to compare work responsibilities semantically.

Experience Score

Compares extracted years of experience against the JD requirement using ratio-based scoring.

Education Score

Checks whether the candidate meets or exceeds the required degree level.

Semantic Similarity

Performs full-document TF-IDF cosine similarity to capture contextual relevance beyond exact keywords.

3. Techniques Used and Rationale

Technique	Used For	Why Chosen
Regex Parsing	Skill, education, and experience extraction	Fast, deterministic, and explainable
Skill Alias Normalization	Mapping related technologies	Reduces false mismatches
TF-IDF + Cosine Similarity	Semantic matching	Lightweight contextual comparison
Rule-Based Validation	Warning detection	Detects manipulation attempts
Evidence Extraction	Explainability	Shows why a skill was matched

Example Alias Mapping

- node.js → javascript
- numpy, pandas → python
- postgresql → sql

This normalization improves recall and reduces terminology-based mismatches.

4. Test Case Findings

The system was evaluated using multiple resume scenarios.

Test Case	Observation
Strong Match	Correctly identified most required skills and produced high scores
Poor Match	Missing critical skills significantly reduced the score
Ambiguous Match(Moderate)	Detected transferable and partially matching skills
Adversarial Input	Generated warnings for keyword-stuffed or suspicious input

Key Findings

- Deterministic parsing produced stable and reproducible outputs.
- TF-IDF improved contextual matching beyond exact keyword overlap.
- Alias normalization reduced missed skill matches.
- Warning detection improved robustness against manipulation.

5. Known Failure Modes

Although effective within the project scope, the system has several limitations:

- Emerging technologies may not exist in the predefined skill mappings.
- Implicit experience mentioned through employment dates may not always be detected accurately.
- Skill proficiency depth and seniority are difficult to estimate.
- Poorly formatted resumes can reduce extraction accuracy.
- TF-IDF similarity may occasionally overestimate related skills.
- Sophisticated keyword stuffing may partially bypass warning detection.

These limitations are treated as uncertainty areas within the scoring pipeline.

6. What I Would Improve With Two More Weeks

Given additional development time, the following improvements would be implemented:

Technical Improvements

- Advanced LLM integration for semantic reasoning
- Better seniority and expertise detection
- Sentence embedding-based semantic similarity
- Improved experience extraction using employment dates
- OCR support for image-based resumes

Feature Improvements

- Resume improvement recommendations
- Candidate ranking dashboard
- Batch resume processing
- Interactive score visualization

AI Improvements

- Hybrid LLM + deterministic scoring
- Better uncertainty estimation
- Improved contextual skill understanding
- Enhanced semantic matching for emerging technologies

Conclusion

The Resume Scoring System demonstrates how a hybrid AI approach can be used to build a transparent, explainable, and lightweight hiring assistance tool within a short development timeline.

By combining deterministic rule-based extraction, semantic similarity techniques, explainable scoring, and adversarial validation checks, the system delivers reliable and understandable hiring insights while remaining reproducible, cost-effective, and easy to audit.